ARCHITECTURE DOCUMENT

(Micro Garden – Automated Indoor Gardening System)

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1. System Requirements Specification (SyRS)

1.1 System Overview

The Micro Garden is a compact, automated indoor greenhouse system intended for personal use in space-constrained environments (e.g., apartments, dorms). The system monitors and regulates key plant-growth parameters such as air temperature, relative humidity, soil moisture, and light intensity using environmental sensors and closed-loop control. It provides real-time monitoring, historical data visibility, and manual override through a user interface.

1.2 Stakeholders and Users

- Primary Users: individuals seeking a compact automated indoor gardening solution (students, apartment residents, hobbyists).
- Academic Stakeholders: course instructor/supervisor/TA/evaluators assessing technical feasibility, integration, testing, and documentation.

1.3 Operational Environment and Assumptions

Operational Environment

- Intended for typical indoor environments (home/school/lab).

Assumptions

- Required sensors/components are available within budget.
- Team has access to lab tools and workspaces.
- Enclosure conditions can be influenced by selected actuators.

1.4 Functional Requirements (FR)

FR-1

Sensor Measurement (Core Parameters)
The system shall measure air temperature, relative humidity, soil moisture, and light intensity using sensors and record readings for monitoring and control.

FR-2

Autonomous Environmental Control
The system shall automatically actuate subsystems (e.g., lighting, irrigation, and climate-related actuators) based on sensor feedback and configured targets/thresholds.

FR-3

Multi-Parameter Closed-Loop Control
The system shall implement closed-loop control using feedback from at least the four core parameters (temperature, humidity, soil moisture, light).

FR-4

Target Maintenance Performance
The system shall maintain each controlled parameter within ±10% of defined target ranges for at least 70% of a continuous 72-hour autonomous operation period.

FR-5

Scheduled Automation
The system shall support scheduled actions (at minimum: lighting and watering) and execute them reliably without missed events during the defined test cycle.

FR-6

User Interface (Real-Time + Historical Data)
The system shall provide a user interface that displays real-time and historical environmental data for all major subsystems/parameters.

FR-7

Manual Override
The system shall allow the user to manually override automation actions (e.g., forcing irrigation or lighting ON/OFF) while preserving transparency of system state.

FR-8

Alerts / Notification of Missed or Abnormal Events
The system shall provide a visible notification when a scheduled action is missed or when a parameter is outside the configured acceptable range for a sustained period (threshold and duration to be defined). *(Derived from "no missed watering event … without notice" requirement.)*

1.5 Non-Functional Requirements (NFR)

NFR-1

Stability (No Sustained Oscillations)
Closed-loop control behavior shall not exhibit sustained oscillations or instability during representative operating conditions.

NFR-2

Usability (UI Task Success)
The user interface shall achieve at least a 95% task-completion success rate during structured user testing (tasks to be defined, e.g., view status, change setpoint, trigger manual watering).

NFR-3

Reliability During Autonomous Operation
The system shall support continuous autonomous operation for at least 72 hours for the performance evaluation described in FR-4.

NFR-4

Modularity / Maintainability
The design should be modular such that sensors/actuators can be tested as subsystems prior to full integration (supports integration risk mitigation).

NFR-5

Safety (Electrical + Water Safety)
The system shall comply with relevant lab-safe practices for electrical and water proximity (e.g., insulation, secure wiring, leak prevention) and follow lab safety regulations.

1.6 Constraints, Limitations, and Out-of-Scope

Constraints

- Schedule: must meet ELG4912 academic deadlines.
- Budget: limited; project is self-funded.
- Space: must remain compact for indoor and lab environments.
- Safety: must respect electrical and water safety requirements.

Out-of-Scope (Explicit)

- Commercial-grade product design/certification/manufacturing optimization.
- Fully autonomous decision-making with no human supervision.
- Outdoor/industrial/extreme-environment deployment.
- Long-term biological performance evaluation across full plant lifecycle.
- Research-grade, laboratory-instrumentation control systems.

1.7 Open Items (To Be Finalized)

- Exact target ranges (numerical setpoints) for temperature, humidity, soil moisture, and light for the selected plant(s).
- Definition of “acceptable range” and “sustained period” for alerts.
- Exact user-testing tasks used to measure the 95% UI task success metric.
- Detailed sensor sampling/logging rates and data retention period (for historical trends).

2. Conceptual Design

2.1 System Architecture Overview

The Micro Garden system follows a closed-loop cyber-physical architecture consisting of sensing, processing, actuation, communication, and user-interaction subsystems. The goal of the architecture is to continuously monitor environmental conditions and automatically regulate plant-growth parameters through feedback control.

At a high level, the system operates according to the following workflow:

1. Environmental sensors measure physical parameters including soil moisture, air temperature, humidity, and light intensity.
2. Sensor data is transmitted to a microcontroller unit (MCU) which performs data processing and decision-making.
3. Control algorithms compare measured values against predefined target ranges.
4. Based on system feedback, the MCU activates appropriate actuators such as the irrigation pump or grow lights.
5. System data is transmitted to a cloud-based monitoring interface, allowing users to observe system status and manually override automation when required.
6. The updated environmental state is measured again, forming a continuous closed-loop control cycle.

This architecture enables autonomous plant maintenance while maintaining transparency and user supervision.

Major Architectural Subsystems

The Micro Garden system is divided into five primary subsystems:

1. Sensing Subsystem

- Soil moisture sensor
- Temperature and humidity sensor (DHT11/DHT22)

- Light intensity sensor

Responsible for real-time environmental data acquisition.

2. Processing and Control Subsystem

- Microcontroller (Arduino Uno / ESP8266 NodeMCU)
- Control logic implementation
- Threshold comparison and scheduling

Responsible for executing automation and feedback control decisions.

3. Actuation Subsystem

- Water pump (irrigation)
- LED grow light
- Optional airflow or humidity control devices

Responsible for modifying environmental conditions.

4. Communication and Data Management Subsystem

- WiFi connectivity via ESP8266
- Arduino IoT Cloud dashboard
- Data logging and remote monitoring

Responsible for transmitting system data and enabling remote interaction.

5. User Interface Subsystem

- Web/mobile dashboard
- Real-time parameter visualization
- Manual override controls

Allows users to supervise and interact with the system.

2.2 System Block Diagram Description

The Micro Garden system architecture is represented using a high-level block diagram illustrating the interaction between sensing, control, actuation, communication, and user interface components.

At the core of the system is the microcontroller unit (MCU), which acts as the central processing and decision-making element. All subsystems communicate either directly or indirectly through the MCU.

Sensor Inputs

Environmental conditions are measured using multiple sensors connected to the microcontroller:

- Soil Moisture Sensor – measures water content in the soil to determine irrigation requirements.
- Temperature and Humidity Sensor (DHT11/DHT22) – monitors ambient climate conditions inside the enclosure.
- Light Sensor – measures illumination levels affecting plant growth.

These sensors continuously provide analog or digital signals representing the current environmental state.

Processing and Control Block

The microcontroller receives sensor data and executes control logic based on predefined thresholds or scheduled parameters. The controller performs:

- Data acquisition and filtering
- Threshold comparison
- Automation scheduling
- Closed-loop feedback decisions

Control outputs are generated whenever environmental parameters deviate from desired operating ranges.

Actuation Outputs

Based on control decisions, the MCU drives actuator modules through relay or driver circuits:

- Water Pump – activates irrigation when soil moisture falls below the required level.
- LED Grow Light – regulates artificial lighting according to scheduled cycles or measured light intensity.
- *(Optional future expansion: fan or humidity control actuator).*

These actuators directly modify environmental conditions inside the Micro Garden.

Communication and Cloud Interface

The system communicates with an external monitoring platform via WiFi using the ESP8266 module. Sensor data and system states are transmitted to the Arduino IoT Cloud dashboard, enabling:

- Remote monitoring
- Historical data visualization
- Parameter configuration
- Manual override control

User Interaction Loop

Users interact with the system through a web or mobile dashboard. Manual commands issued by the user are transmitted back to the MCU, temporarily overriding autonomous control when required.

Closed-Loop Operation

The architecture forms a continuous feedback loop:

Sensor Measurement → Controller Decision → Actuator Response → Environmental Change → Updated Measurement

This loop enables autonomous regulation of plant-growth conditions while maintaining user supervision.

2.3 Hardware Architecture

The hardware architecture of the Micro Garden system is designed to enable reliable environmental sensing, autonomous control, and safe actuation within a compact indoor enclosure. The system integrates sensing devices, processing hardware, actuator modules, communication components, and power management circuitry.

2.3.1 Processing Unit

The central processing element of the system is a microcontroller platform consisting of an Arduino Uno and an ESP8266 NodeMCU module.

- The Arduino Uno is responsible for sensor interfacing and actuator control.
- The ESP8266 NodeMCU provides WiFi connectivity and enables communication with the Arduino IoT Cloud platform.

The microcontroller performs:

- Sensor data acquisition
- Control decision execution
- Automation scheduling
- Communication with the user interface

This distributed configuration separates real-time control from wireless communication tasks, improving system stability.

2.3.2 Sensor Subsystem

Environmental monitoring is achieved using multiple sensors connected to the microcontroller input pins:

- Soil Moisture Sensor
  - Measures soil water content using analog voltage output.
  - Connected to an analog input pin of the Arduino.
  - Used to determine irrigation requirements.
- Temperature and Humidity Sensor (DHT11)
  - Provides digital measurements of ambient temperature and relative humidity.
  - Connected to a digital GPIO pin.
  - Supports climate monitoring within the enclosure.
- Light Intensity Sensor
  - Measures available light levels affecting plant growth.
  - Provides feedback for lighting automation decisions.

These sensors continuously supply environmental feedback required for closed-loop control operation.

2.3.3 Actuation Subsystem

Actuators modify environmental conditions based on controller outputs.

- Water Pump
  - Activated when soil moisture falls below the defined threshold.
  - Controlled through a relay or transistor driver circuit to safely switch higher-current loads.
- LED Grow Light
  - Provides artificial illumination for plant growth.
  - Operates according to scheduled timing or measured light intensity.

Driver circuitry electrically isolates the microcontroller from actuator power loads, ensuring safe operation.

2.3.4 Communication Hardware

Wireless communication is implemented using the ESP8266 WiFi module, enabling:

- Transmission of sensor data to the Arduino IoT Cloud dashboard
- Remote monitoring of environmental conditions
- Reception of user commands and manual overrides

This communication layer enables real-time supervision without direct physical interaction with the system.

2.3.5 Power Architecture

The system utilizes separate regulated power paths to ensure stable operation:

- 5V supply for Arduino and sensor modules
- External supply for actuator loads such as the water pump and lighting system

Power separation prevents voltage drops and electrical noise generated by actuators from affecting sensor measurements or controller stability.

Proper grounding and insulation practices are implemented to maintain electrical safety, particularly due to proximity between water and electronic components.

2.3.6 Hardware Integration Strategy

The hardware architecture follows a modular integration approach:

1. Individual sensor validation
2. Actuator driver testing
3. Microcontroller integration
4. Communication verification
5. Full-system closed-loop testing

This staged integration reduces debugging complexity and minimizes system integration risks.

2.4 Software Architecture

The Micro Garden software architecture is designed to enable autonomous environmental monitoring, closed-loop control, cloud communication, and user interaction. The software system operates using an event-driven and periodically sampled control structure implemented on the microcontroller platform.

2.4.1 Software Structure Overview

The software system is divided into four primary functional layers:

1. Sensor Data Acquisition Layer
2. Control and Decision Layer
3. Communication Layer

4. User Interaction Layer

Each layer performs a specific function while interacting through shared system variables.

2.4.2 Sensor Data Acquisition Layer

The data acquisition layer periodically reads environmental measurements from connected sensors:

- Soil moisture sensor readings obtained through analog input sampling.
- Temperature and humidity measurements collected using the DHT sensor library.
- Light intensity values measured using analog or digital sensor input.

Sensor readings are sampled at fixed time intervals and stored as system state variables used by the control algorithm.

Basic validation and filtering are applied to reduce noise and prevent erroneous actuator activation.

2.4.3 Control and Decision Layer

The control layer implements the system's closed-loop automation logic.

Measured environmental parameters are continuously compared against predefined target thresholds. When deviations are detected, control decisions are generated according to programmed rules.

Examples include:

- Activating irrigation when soil moisture falls below threshold.
- Enabling grow lights when illumination is insufficient or during scheduled periods.
- Maintaining acceptable environmental operating ranges.

The controller operates using threshold-based feedback control combined with scheduled automation routines.

2.4.4 Scheduling and Automation Logic

Time-based automation is implemented using internal timers or cloud-synchronized scheduling.

Scheduled operations include:

- Lighting cycles simulating day/night conditions.
- Periodic irrigation events.

Automation routines execute independently of user interaction, enabling fully autonomous operation during extended periods.

This scheduling mechanism ensures reliable execution of actions without missed events.

2.4.5 Communication and Cloud Integration

The system communicates with the Arduino IoT Cloud platform via the ESP8266 WiFi module.

Software responsibilities include:

- Uploading real-time sensor data
- Synchronizing device variables with cloud dashboards
- Receiving remote commands from users
- Updating actuator states based on dashboard inputs

Cloud synchronization allows remote supervision and data visualization without direct physical access to the system.

2.4.6 User Interface Interaction

Users interact with the Micro Garden through a web or mobile dashboard.

The interface allows users to:

- View live environmental conditions
- Observe historical trends
- Modify control parameters
- Manually override automated actions

Manual commands temporarily supersede autonomous control logic while maintaining system monitoring functionality.

2.4.7 Main Control Loop Operation

The Micro Garden software operates using a continuous execution loop structured as follows:

1. Read sensor values
2. Update system variables

3. Evaluate control conditions
4. Activate or deactivate actuators
5. Transmit updated data to cloud dashboard
6. Check for user commands

This loop repeats continuously, forming the software implementation of the closed-loop control system.

2.5 Control Strategy

The Micro Garden system implements a closed-loop environmental control strategy designed to automatically regulate plant-growth conditions using continuous sensor feedback and actuator adjustment.

2.5.1 Control Approach

The system utilizes a threshold-based feedback control method. Environmental parameters measured by sensors are continuously compared against predefined target ranges. When a measured parameter deviates outside the acceptable range, corrective action is applied through system actuators.

Examples of control actions include:

- Activating the irrigation pump when soil moisture falls below the required threshold.
- Enabling grow lights when light intensity is insufficient or during scheduled operating periods.
- Maintaining acceptable temperature and humidity conditions through monitored supervision.

Once environmental conditions return to acceptable limits, actuators are automatically deactivated.

2.5.2 Closed-Loop Feedback Operation

The Micro Garden operates as a feedback control system consisting of:

- Plant Environment (Process): soil and air conditions affecting plant growth
- Sensors (Measurement): environmental parameter monitoring
- Controller: microcontroller executing decision logic
- Actuators: pump and lighting system modifying conditions

The resulting feedback loop can be summarized as:

Measurement → Comparison → Control Decision → Actuation → Environmental Response → Updated Measurement

This continuous loop enables autonomous regulation without constant user intervention.

2.5.3 Stability Considerations

A threshold-based control strategy was selected due to the slow dynamics of plant-growth environments. Changes in soil moisture, temperature, and lighting occur gradually, reducing the likelihood of rapid oscillations or instability.

To further improve stability:

- Sensor readings are periodically sampled rather than continuously switched.
- Actuators operate only when thresholds are exceeded.
- Control actions are limited to discrete ON/OFF responses.

These characteristics prevent excessive actuator switching and ensure stable environmental regulation.

2.5.4 Manual Override and Supervisory Control

Although autonomous control is the primary operating mode, users may manually override actuator states through the cloud dashboard. Manual commands temporarily supersede automated decisions while maintaining system monitoring functionality.

This supervisory control structure ensures both automation reliability and user authority over system operation.

2.5.5 Control Strategy Justification

The selected control approach provides several advantages for the Micro Garden application:

- Low computational complexity suitable for embedded microcontrollers
- Reliable operation under slow environmental dynamics
- Easy tuning and implementation
- Reduced power consumption
- High robustness for small-scale indoor systems

Given the project scope and operational requirements, threshold-based feedback control provides an effective balance between performance, simplicity, and reliability.

3. Hardware–Software Interface Definition

The Micro Garden system relies on well-defined interfaces between hardware components and software control modules. These interfaces enable reliable communication between sensors, actuators, processing units, and cloud services.

Clearly defining interfaces improves system modularity, integration, and troubleshooting during implementation.

3.1 Sensor Interfaces

Component	Interface Type	MCU Connection	Data Type	Software Function
Soil Moisture Sensor	Analog Input	Arduino Analog Pin (A0)	Analog Voltage	Reads soil moisture level for irrigation control
DHT11 Temperature & Humidity Sensor	Digital GPIO	Arduino Digital Pin	Digital Data	Provides temperature and humidity measurements
Light Sensor	Analog Input	Arduino Analog Pin	Analog Voltage	Measures environmental illumination

Sensor interfaces convert physical environmental conditions into electrical signals that are periodically sampled by the software acquisition layer.

3.2 Actuator Interfaces

Component	Interface Type	Control Method	Output Signal	Software Role

Water Pump	Relay / Driver Circuit	Digital Output	ON/OFF Signal	Activates irrigation system
LED Grow Light	Relay / MOSFET Driver	Digital Output	ON/OFF Signal	Controls lighting cycle

Actuators are controlled using digital output signals generated by the microcontroller based on control decisions.

3.3 Communication Interface

Component	Interface Type	Protocol	Purpose
ESP8266 NodeMCU	Serial / WiFi	UART / TCP-IP	Cloud communication and remote monitoring
Arduino IoT Cloud	Wireless Network	MQTT / HTTP	Data synchronization and user commands

The communication interface enables bidirectional data exchange between the embedded controller and cloud dashboard.

3.4 User Interface Interaction

Interface Element	Direction	Function
Dashboard Display	System → User	Shows real-time and historical environmental data
Manual Override Controls	User → System	Sends actuator control commands

Parameter Settings	User → System	Updates automation thresholds

User commands received through the cloud platform are processed by the software control layer and translated into hardware actions.

3.5 Interface Design Considerations

The interface architecture was designed to ensure:

- Electrical isolation between control signals and high-power actuators
- Reliable sensor sampling without communication interference
- Modular subsystem replacement without major software modification
- Safe operation in environments involving water and electronics

These interface definitions support scalable system integration and simplify debugging during testing and deployment.

2.6 System Architecture Diagram

Figure 2.1 illustrates the overall system architecture of the Micro Garden, showing the interaction between sensing components, processing units, actuators, communication modules, and the user interface.

The architecture follows a closed-loop cyber-physical system structure in which environmental data is continuously measured, processed, and used to regulate plant-growth conditions.

Diagram Description

The system architecture is organized into five major functional blocks:

1. Sensor Layer Environmental sensors collect real-world data from the plant enclosure:

- Soil Moisture Sensor
- Temperature and Humidity Sensor (DHT11)
- Light Intensity Sensor

These sensors convert physical environmental conditions into electrical signals transmitted to the controller.

2. Processing and Control Layer The Arduino Uno microcontroller acts as the central controller responsible for:

- Sensor data acquisition
- Control decision execution
- Automation scheduling
- Actuator command generation

Control decisions are produced by comparing measured values with predefined environmental thresholds.

3. Actuation Layer Control signals generated by the microcontroller activate actuator subsystems:

- Water Pump (irrigation control)
- LED Grow Light (lighting regulation)

These actuators directly influence plant environmental conditions.

4. Communication Layer The ESP8266 NodeMCU enables wireless communication between the embedded system and the Arduino IoT Cloud platform. This layer supports:

- Real-time data transmission
- Remote monitoring
- Reception of user commands

5. User Interface Layer Users interact with the system through a cloud dashboard that allows:

- Visualization of environmental parameters
- Historical data monitoring
- Manual override of automation functions

Closed-Loop Interaction

The architecture forms a continuous feedback loop:

Sensors → Controller → Actuators → Plant Environment → Sensors

This feedback mechanism enables autonomous regulation while maintaining user supervision through remote access.

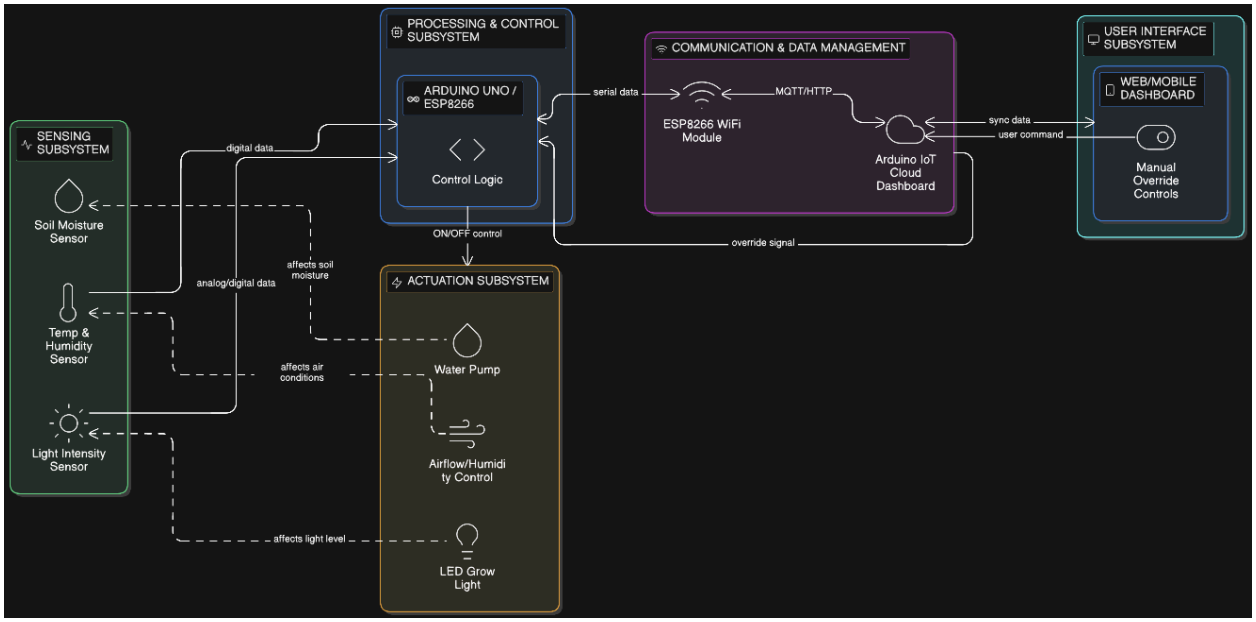

Figure 2.1 — Micro Garden System Architecture Diagram

4. Design Rationale and Tradeoffs

The Micro Garden architecture was designed to balance system performance, implementation complexity, cost constraints, and reliability within an academic prototyping environment. Key architectural decisions were made based on project scope, available resources, and operational requirements.

4.1 Microcontroller Selection

The Arduino Uno was selected as the primary controller due to its reliability, extensive library support, and compatibility with environmental sensors and actuator modules. Its simplicity enables rapid prototyping and debugging while providing sufficient processing capability for threshold-based control.

The ESP8266 NodeMCU was incorporated to provide wireless connectivity and cloud communication. Separating control execution and communication tasks improves system stability and prevents network operations from interrupting real-time control processes.

4.2 Control Strategy Selection

A threshold-based feedback control strategy was chosen instead of advanced controllers such as PID control.

This decision was motivated by:

- Slow environmental dynamics of plant-growth systems
- Reduced computational requirements
- Easier tuning and implementation
- Improved robustness for discrete actuator operation

Since soil moisture, temperature, and lighting change gradually, ON/OFF regulation provides adequate performance without introducing instability or unnecessary system complexity.

4.3 Cloud-Based Monitoring Choice

Arduino IoT Cloud was selected to enable remote monitoring and interaction without developing a custom backend infrastructure.

Advantages include:

- Rapid dashboard deployment
- Built-in device synchronization
- Remote accessibility
- Data visualization capabilities

This approach allows focus on embedded system integration rather than web infrastructure development.

4.4 Modular Hardware Architecture

A modular design philosophy was adopted to allow independent subsystem testing prior to full integration.

Benefits include:

- Simplified debugging
- Reduced integration risk
- Component replacement flexibility
- Scalable future expansion

This approach supports incremental development consistent with academic project timelines.

4.5 Design Tradeoffs

Several tradeoffs were considered during system design:

- Threshold control provides simplicity but lower precision compared to continuous controllers.
- Separate microcontrollers increase hardware complexity but improve communication reliability.
- Cloud connectivity improves usability but introduces network dependency.

These tradeoffs were accepted to prioritize reliability, implementation feasibility, and successful prototype demonstration.

5. Conclusion

This architecture document presented the system design for the Micro Garden automated indoor gardening system. The proposed architecture integrates environmental sensing, embedded control, wireless communication, and user interaction into a unified closed-loop cyber-physical system.

The hardware and software architectures enable autonomous regulation of plant-growth conditions while maintaining user supervision through cloud-based monitoring and manual override capabilities. Clearly defined subsystem interfaces and modular integration strategies support reliable implementation and future scalability.

The selected architectural approach balances performance, simplicity, and feasibility within project constraints, providing a practical foundation for system implementation, testing, and validation in subsequent project phases.